

# **Atlantic Sea Scallop Management History and Current Fishery Overview**

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## **1. Species Overview**

The Atlantic sea scallop (*Placopecten magellanicus*) is a bivalve mollusk found in the Northwest Atlantic Ocean, from Newfoundland, Canada to Cape Hatteras, North Carolina. Atlantic sea scallops are generally found at depths of about 100 to 300 feet (30 to 90 meters) on Georges Bank and in the Mid-Atlantic, though they can be found in shallower waters in the Gulf of Maine and Canada. They can live up to 20 years, with rapid growth occurring during the first few years of life. Scallops feed by filtering phytoplankton or other small organisms out of the water column. While they are predominately a sedentary species, sea scallops can swim by using their adductor muscle to snap the top and bottom shells open and shut. This can propel them through the water and is a behavior that is used to escape predators. Atlantic sea scallops are highly fertile broadcast spawners that can reproduce by age 2, though they do not produce large amounts of eggs or sperm until they are four years old. Fully mature females can produce hundreds of millions of eggs per year.

## **2. History of Atlantic Sea Scallop Management**

The U.S. Atlantic sea scallop fishery occurs from the Maine/Canada border to Ocean City, Maryland, and from inshore to offshore waters on the edge of the continental shelf (Map 1). The fishery predominantly uses dredges, and to a lesser extent, bottom trawls in the Mid-Atlantic. Scallops are processed by hand at sea, with the abductor muscle retained for human consumption. Additional information about scallop management and landings are contained in TOR 2.

The New England Fishery Management Council (NEFMC) and NOAA Fisheries/National Marine Fisheries Service (NMFS) manage the fishery in federal waters, while Maine and Massachusetts manage smaller scale inshore fisheries occurring in their state waters.

The Scallop Fishery Management Plan (FMP) was established in 1982 to address the overall long-term benefits from the harvest and use of the sea scallop resource. The plan has been modified considerably over time through a series of Amendments and

Framework Adjustments. Key management actions include license limitations in 2004 (Amendment 4) and again in 2010 in conjunction with a catch share program (Amendment 11), effort controls (e.g., days-at-sea), and rotational management of productive scallop beds to improve yield per recruit (Amendment 10). Amendment 15 to the FMP was developed to comply with the 2007 revisions to the Magnuson-Stevens Act, and implemented a new harvest control rule and annual catch limits. Other important management measures include a minimum ring size of 4", minimum mesh size of 10" on the top of the dredge to allow bycatch to swim out if captured, and a seven-person crew size limit.

### ***Federal Management Actions***

The NEFMC makes proposals, through various management actions, to NMFS on the management of the fishery. The federal FMP has been modified through several subsequent Amendment, Framework Adjustment, and Specifications actions over the years. Key actions by year are described briefly in the bullets below.<sup>1</sup>

- 1982 – The Atlantic Sea Scallop Fishery Management Plan implemented.
- 1994 – NOAA Fisheries closed areas on Georges Bank and Nantucket shoals to groundfish and scallop fishing.
- 1994 - Moratorium on scallop permits implemented; limits on "days-at-sea" (DAS) to reduce fishing effort and a limit on crew size initiated; protocol for collecting commercial fishing data for Northeast U.S. fisheries changed; increase in dredge ring size from 3.0 to 3.5 inches required by 1996; open access "general category" scallop permits created to allow small-scale scalloping and incidental catch of scallops in other fisheries (Amendment 4).
- 1997 – Scallops declared overfished.
- 1998 - More stringent DAS limitation and a plan to rebuild stocks within 10 years established; maximum crew size capped at seven; vessel monitoring systems required on limited access scallop vessels.
- 1998 – 2001 – Managers closed large areas of the Mid-Atlantic to scallop fishing.
- 1999 – 2001 – First limited re-openings of closed areas of Georges Bank to directed scallop fishing. Precursor to the full development and adoption of rotational management.
- 2004 - Area rotation management program implemented (rotating open and closed areas to maximize scallop yield); dredge ring size was further increased to 4 inches; minimum mesh size for mesh on top of the dredge gear set at 10 inches to reduce flounder bycatch (Amendment 10).

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<sup>1</sup> See also: <https://www.nefmc.org/management-plans/scallops>

- 2006 - Chain mats required in sea scallop dredges used in the Mid-Atlantic, May through November, to prevent sea turtles from entering gear.
- 2008 - Limited access (LA) and individual quotas established for the general category fleet of scallop vessels; the limited access general category (LAGC) scallop fishery was created and there was no longer an open access scallop permit. There were three categories of LAGC permits: Individual fishing quota (IFQ) permits, Northern Gulf of Maine (NGOM) permits, and incidental permits.
- 2011 - Annual catch limit (ACL) and accountability measure (AM) requirements of the Magnuson-Stevens Fishery Conservation and Management Act implemented, including a sub-ACL and AM for yellowtail flounder caught in the scallop fishery; The Council adopted an ACL structure for the scallop FMP where the overfishing limit (OFL) is greater than Acceptable Biological Catch (ABC); ABC is equal to overall ACL; and overall ACL is greater than Annual Catch Target (ACT; represented as  $OFL > ABC = ACL > ACT$ ). Scallop Essential Fish Habitat (EFH) closed areas were modified to be consistent with areas closed for EFH in the Northeast Multispecies Fishery Management Plan (Amendment 15).
- 2016 – Amendment 19 changed the start of the fishing year from March 1 to April 1 and established a streamlined specifications setting process.
- 2018 – Revisions to habitat closures reopened scallop grounds on inshore Georges Bank and south of Nantucket, spurring an increase in fishery landings.
- 2022 – In response to a growing directed fishery in the NGOM management unit, Amendment 21 established a new allocation sharing arrangement to support research, monitoring, and a directed fishery with trips limits for LAGC NGOM and IFQ vessels below a biomass threshold. The amendment also implemented provisions to enable all federally permitted scallop vessels to fish in the NGOM when/if biomass reaches higher levels; Increased the LAGC IFQ possession limit in open rotational areas (access areas) to 800 pounds per trip.
- 2024 – Managers increased the frequency of vessel positioning (using VMS) from 30-minutes to 5-minutes.

### **3. Overview of Current Management**

*Fishery Participation.* There are currently two permit types in the federally managed scallop fishery (Limited Access and Limited Access General Category), with multiple permit categories for each type.

In 1994 (Amendment 4), a Limited Access program was created. Limited Access vessels were assigned different DAS limits according to which permit category they

qualified for: full-time, part-time, or occasional. Amendment 4 also created the General Category scallop permit for vessels that did not qualify for a Limited Access permit. Although originally created for an incidental catch of scallops in other fisheries, and for small-scale directed fisheries, the General Category fishery and fleet evolved after its creation in 1994.

The General Category scallop fishery was established as an “open access” fishery, any vessel that wanted to apply for a permit could; there were no specific qualifications to receive a general category permit. The main control on mortality for this component of the scallop fishery was a daily possession limit. Amendment 11, implemented in 2008, transitioned the general category component from an open access fishery to limited access. Vessels with at least 1,000 lb of landings history during a qualifying year (2000 – 2004) were eligible for an IFQ permit and “contribution factor” (allocation), while general category vessels that did not qualify for an IFQ permit were eligible for a NGOM scallop permit, or an incidental catch permit.

Since 2008, all federal scallop permits have been limited access. A vessel can hold LA permits only, LAGC permits only, or a combination of LA and LAGC permits. There are multiple permit categories within LA and LAGC (Table 1). As of 2023, there were around 340 active Limited Access vessels, 90 active LAGC IFQ vessels, and 130 active LAGC NGOM vessels.<sup>2</sup>

*Stock Status.* The Atlantic sea scallop resource was determined to be not overfished, and overfishing was not occurring at the 2020 management track stock assessment. An overview of past reference points and stock status determinations is shown in Table 2.

The NEFMC generally sets specifications for the fishery on an annual basis. This process includes designations of OFL, ABC, ACLs, and ACTs for the scallop fishery, as well as scallop catch for the NGOM, incidental, and state waters catch components of the scallop fishery. The scallop fishery assessments determine the exploitable biomass, including an assessment of discard and incidental mortality (mortality of scallops resulting from interaction, but not capture, in the scallop fishery).

*Overfishing Limit.* The OFL is specified as the level of catch and associated fishing mortality rate ( $F$ ) that, above which, overfishing is occurring. The OFL will account for landings of scallops in state waters by vessels without Federal scallop permits. The 2020 stock assessment (NEFSC 2020) set the OFL where  $F = 0.61$ , which is the average of the Georges Bank and Mid-Atlantic as they are modeled separately.

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<sup>2</sup> See Framework 39 for more details: <https://www.nefmc.org/library/scallop-framework-39>

*Acceptable Biological Catch/Annual Catch Limit.* The NEFMC's Atlantic sea scallop ABC control rule is biomass-based. Implemented through Amendment 15 to the Atlantic Sea Scallop FMP, the ABC control rule has been in place since 2011.

When applied, the ACL is equal to the ABC in the Scallop FMP. To account for scientific uncertainty, ABC is set at the F that has a 25% probability of exceeding the F associated with OFL (i.e., a 75% probability of being below the F associated with the OFL). The 2020 management track assessment determined that the F associated with the ABC/ACL is  $F=0.45$ . As specified in Amendment 21, exploitable biomass from the Northern Gulf of Maine contributes to the overall OFL and ABC. Observer and research set-asides are removed from the ABC (1% of the ABC/ACL and 1.275 mil lb. (578 mt), respectively). The NGOM Set-Aside, which is available for directed LAGC fishing, is also removed before calculating the legal limits for LA and LAGC IFQ. The remaining available landings (allocation) are divided between the LA and LAGC fisheries into two sub-ACLs: 94.5% for the LA fishery sub-ACL, and 5.5% for the LAGC fishery sub-ACL. Figure 1 summarizes how the various ACL terms are related in the Scallop FMP.

*Annual Catch Targets.* For each sub-ACL there is an ACT to account for management uncertainty. For the LA fleet, the ACT has an associated 75% probability that the ACT will not exceed the ABC/ACL. The F associated with the LA ACT is  $F = 0.39$ . The major sources of management uncertainty in the LA fishery are carryover provisions including the 10 DAS carryover provision and allowing vessels to fish unused access area allocation from the previous fishing year within the first 60 days of the fishing year. For the LAGC fleet, the ACT is equal to the LAGC fleet's sub-ACL, since this component is managed entirely by quotas and is presumed to have less management uncertainty. The fishery specifications allocated to the fishery may be set at an F rate lower than the ACT, but fishery specifications may not exceed this level.

*Annual Projected Landings.* The annual projected landings (APL) are developed using a forecasting model (Scallop Area Management Simulator or SAMS) of the scallop resource that is available for harvest (open areas and available access areas). The APL combines projected landings of exploitable scallops from open area DAS when fishing at an F determined by the Council and expected landings from access areas. While the OFL and ABC include scallops in areas closed to the fishery, the APL is the value of expected harvest and is not contained in the ACL flowchart (Figure 1). This value is consistently below the legal limits of the fishery (OFL, ABC, ACL) as shown in Figure 2. The APL is allocated between the Limited Access component (94.5%) and the LAGC IFQ component (5.5%).

*Rotational Management.* The NEFMC formally adopted a rotational management program in Amendment 10. When sufficient densities of juvenile scallops are detected

in a large enough area, the area is closed to directed scallop fishing to allow the young scallops to grow larger (improve yield per recruit) and to reproduce. These temporary closures can reduce bycatch on non-target species. After multiple years of closure, the areas re-open to directed scallop fishing. Full-time LA vessels receive individual trip allocations, and the LAGC fleet is allocated a total number of trips into open rotational areas that they can fish.

*Bycatch and Accountability Measures.* Other species are caught incidentally in the directed Atlantic sea scallop fishery. The species composition varies based on season, and area, but some of the species caught include yellowtail flounder (*Limanda ferruginea*), windowpane flounder (*Scophthalmus aquosus*), skate species, and monkfish (*Lophius americanus*). Catches are sorted on deck by fishing crews before scallops are processed.

The scallop fishery is allocated sub-ACLs of Georges Bank yellowtail flounder, and Gulf of Maine / Georges Bank windowpane flounder, Southern New England yellowtail flounder, and Southern New England / Mid Atlantic windowpane flounder. The fishery is required to use modified gear in specific areas (AMs) if / when sub-ACLs are exceeded.

Map 1 - Atlantic sea scallop management areas in the Gulf of Maine, Georges Bank, and the Mid-Atlantic for Fishing Year 2023. Map Source: NOAA Fisheries.

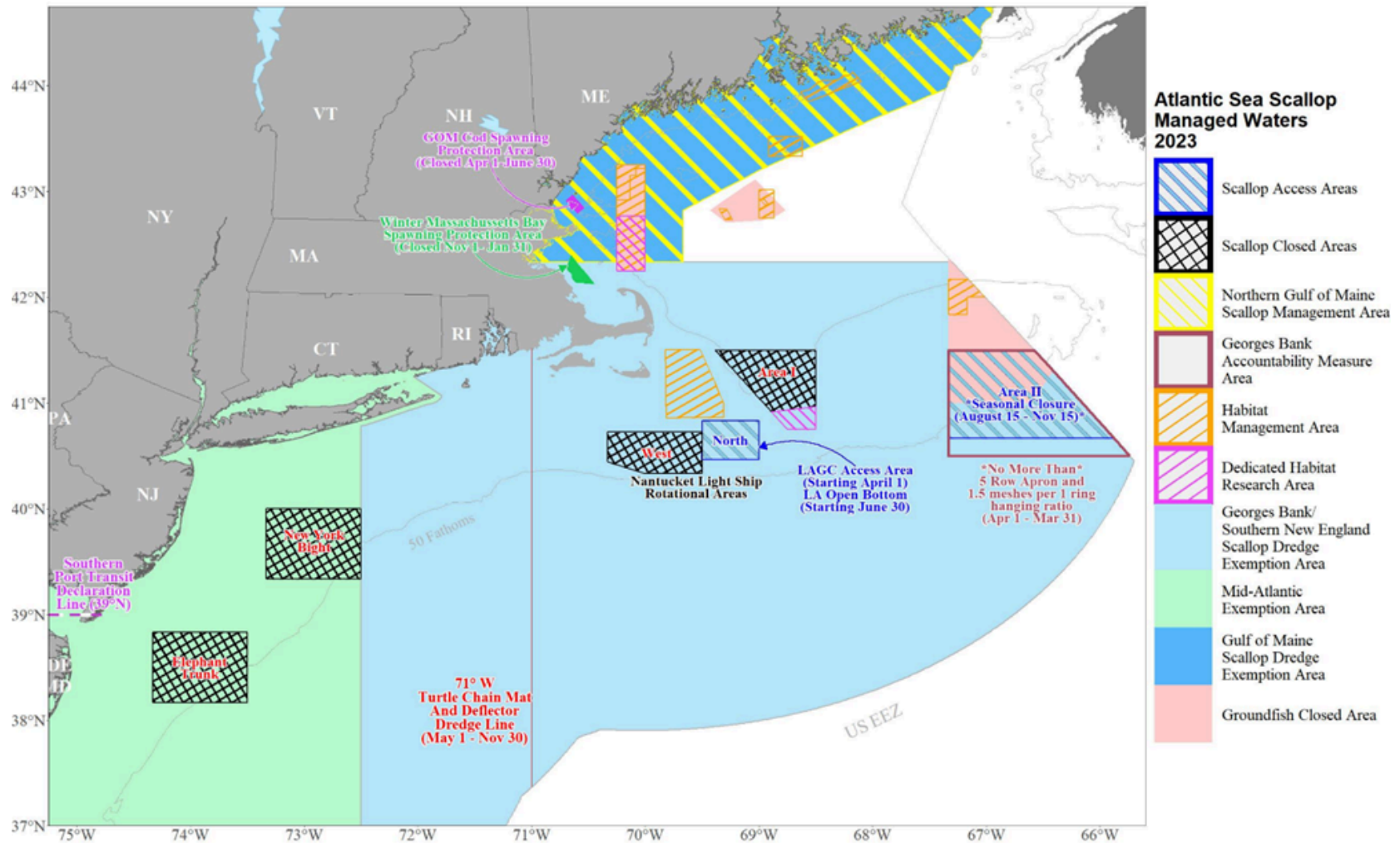


Table 1 - Overview of Scallop Permit Types and Categories.

| Permit Type                                                                                                       |                   | Year Created | Action    | Qualifying Criteria                                                                               | Permit Category                                                                                                                     | Harvest Limits                                                               | Vessel level allocation? | Form of allocation                                 |
|-------------------------------------------------------------------------------------------------------------------|-------------------|--------------|-----------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------|----------------------------------------------------|
| <b>Limited Access<sup>a</sup></b>                                                                                 |                   | 1994         | Amend. 4  | One trip with over 400 pounds in either 1988 or 1989, extended for new vessels under construction | Based on number of days used in 1990, or average of 1985-1990 days                                                                  | 94.5% of APL, after set-asides and incidental catch removed                  | Yes                      | DAS and access area trips                          |
| <b>LA General Category</b>                                                                                        | <b>IFQ</b>        | 2008         | Amend. 11 | Possess Open Access GC permit                                                                     | 1,000 pounds landings in a year (FY2000-2004), individual allocation based on best year indexed by # of years active in the fishery | 5.5% of APL, after set-asides and incidental catch removed                   | Yes                      | IFQ pounds; set # AA trips at fleet level          |
|                                                                                                                   | <b>NGOM</b>       | 2008         | Amend. 11 | Possess Open Access GC permit                                                                     | No landings history required                                                                                                        | Up to TAL for management area, specific landings limits based on survey data | No                       | Harvest in area until LAGC fleet reaches set-aside |
|                                                                                                                   | <b>Incidental</b> | 2008         | Amend. 11 | Possess Open Access GC permit                                                                     | No landings history required                                                                                                        | Deducted from APL before allocating to LA and LAGC IFQ                       | No                       | Harvest allowed until limit is reached             |
| <i>Note: There are multiple categories of LA permits (full-time/part-time, dredge/trawl, small/large dredge).</i> |                   |              |           |                                                                                                   |                                                                                                                                     |                                                                              |                          |                                                    |



Table 2 - Atlantic sea scallop reference points and status determination from previous stock assessments.

|                       | Definition in Scallop FMP            | SARC 50 (2010) | SARC 59 (2014) | SARC 65 (2018) | 2020 Management Track |
|-----------------------|--------------------------------------|----------------|----------------|----------------|-----------------------|
| OFL                   | $F_{MSY}$                            | $F=0.38$       | $F=0.48$       | $F=0.64$       | $F=0.61$              |
| ABC=ACL               | 25% probability of exceeding the OFL | $F=0.32$       | $F=0.38$       | $F=0.51$       | $F=0.45$              |
| $B_{MSY}$             | $B_{TARGET}$                         | 125,358 mt     | 96,480 mt      | 116,766 mt     | 102,657 mt            |
| $\frac{1}{2} B_{MSY}$ | $B_{THRESHOLD}$                      | 62,679 mt      | 48,240 mt      | 58,383 mt      | 51,329 mt             |
| MSY                   |                                      | 24,975 mt      | 23,798 mt      | 46,531 mt      | 32,079 mt             |
| Overfished?           | $B < B_{THRESHOLD}$                  | No             | No             | No             | No                    |
| Overfishing?          | $F < F_{THRESHOLD}=F_{MSY}$          | No             | No             | No             | No                    |

Figure 1 - Overview of annual catch limit flow chart for Atlantic sea scallops.

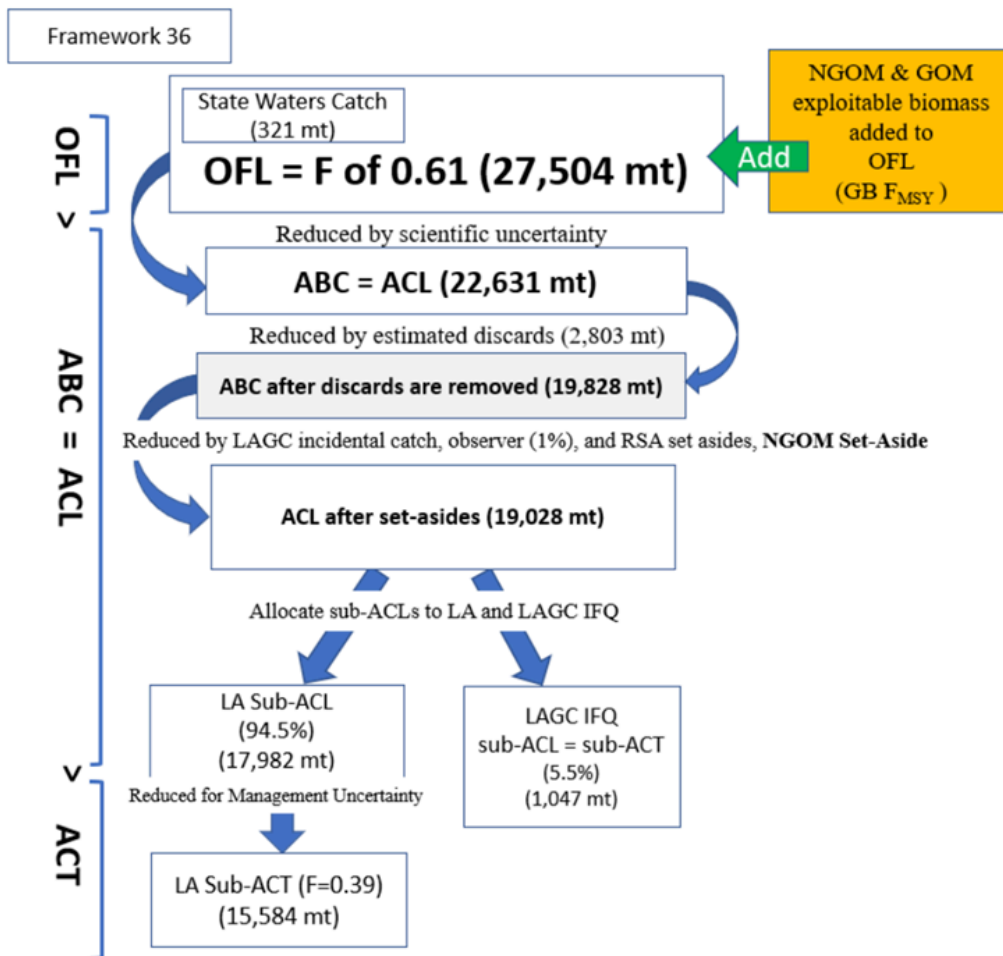
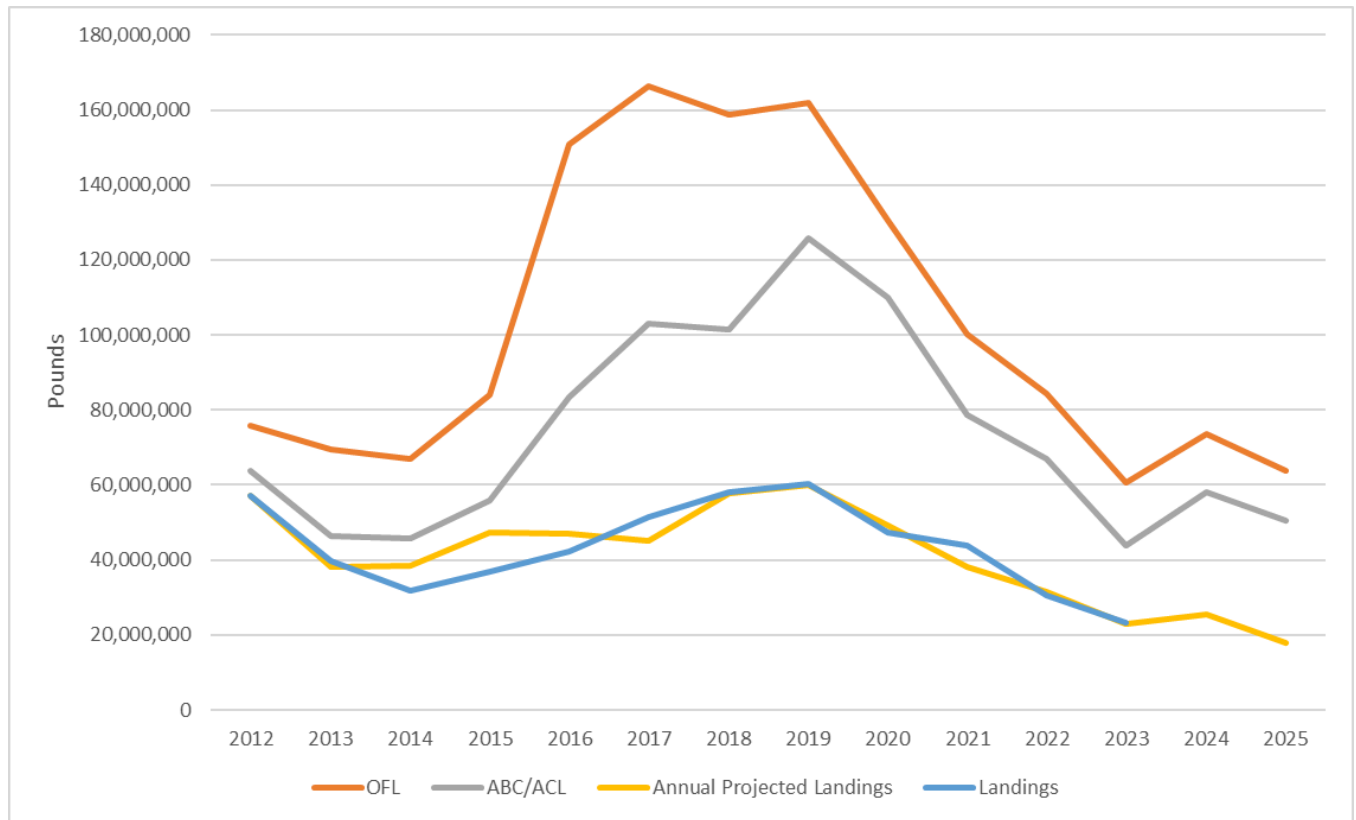


Figure 2 - Overfishing limit, Acceptable Biological Catch/Annual Catch Limit, Annual Projected Landings, and Landings for fishing years 2012 - 2025.



*Table 3 - Atlantic sea scallop landings by fleet component, total revenues, and price per pound for fishing years 2009 - 2023.*

| FY   | Sea Scallop Landings (pounds) |            |                | Total Revenues |                        | Price per pound        |                       |
|------|-------------------------------|------------|----------------|----------------|------------------------|------------------------|-----------------------|
|      | LAGC                          | LA         | Total Landings | Nominal \$     | Real \$<br>(in 2023\$) | Real \$<br>(in 2023\$) | Price<br>(Nominal \$) |
| 2009 | 3,835,951                     | 52,226,623 | 56,062,574     | \$371,819,179  | \$534,991,624          | \$9.20                 | \$6.39                |
| 2010 | 2,165,434                     | 53,294,499 | 55,459,933     | \$458,118,847  | \$645,237,813          | \$11.38                | \$8.08                |
| 2011 | 2,880,337                     | 54,995,259 | 57,875,596     | \$583,971,755  | \$798,866,970          | \$13.59                | \$9.93                |
| 2012 | 2,899,869                     | 52,978,655 | 55,878,524     | \$556,885,194  | \$746,494,898          | \$13.11                | \$9.78                |
| 2013 | 2,364,044                     | 36,981,446 | 39,345,490     | \$460,650,831  | \$610,942,747          | \$15.21                | \$11.47               |
| 2014 | 2,179,696                     | 29,543,485 | 31,723,181     | \$404,214,369  | \$536,805,271          | \$16.57                | \$12.48               |
| 2015 | 2,497,660                     | 32,895,895 | 35,393,555     | \$439,278,060  | \$577,997,447          | \$16.09                | \$12.23               |
| 2016 | 3,616,411                     | 37,116,682 | 40,733,093     | \$495,082,052  | \$633,907,877          | \$15.27                | \$11.92               |
| 2017 | 2,705,736                     | 49,949,005 | 52,654,741     | \$524,525,216  | \$656,477,116          | \$12.26                | \$9.80                |
| 2018 | 3,048,919                     | 56,452,974 | 59,501,893     | \$556,489,739  | \$683,648,328          | \$11.37                | \$9.25                |
| 2019 | 2,848,513                     | 56,818,128 | 59,666,641     | \$561,101,459  | \$679,299,587          | \$11.24                | \$9.29                |
| 2020 | 2,717,609                     | 42,672,438 | 45,390,047     | \$481,738,995  | \$568,088,437          | \$12.36                | \$10.48               |
| 2021 | 2,255,819                     | 39,717,061 | 41,972,880     | \$684,342,262  | \$743,850,285          | \$17.41                | \$16.02               |
| 2022 | 2,422,145                     | 27,527,956 | 29,950,101     | \$434,460,034  | \$449,751,588          | \$14.77                | \$14.27               |
| 2023 | 1,694,816                     | 23,339,033 | 25,033,849     | \$324,055,049  | \$324,055,049          | \$12.76                | \$12.76               |

Table 4 - Summary of allocated DAS and Access Area Trips from 2010 - 2024.

| Year+             | Action      | DAS   | AA trips | CA I              | CA II     | NLS                       | HC                                                              | ETA                       | DMV                                   | NYB    | Poss. Limit |
|-------------------|-------------|-------|----------|-------------------|-----------|---------------------------|-----------------------------------------------------------------|---------------------------|---------------------------------------|--------|-------------|
| 2010              | FW21        | 38    | 4        | Closed            | Closed    | 1 trip                    | Closed                                                          | 2 trips                   | 1 trip                                |        | 18,000      |
| 2011              | FW22 and EA | 32    | 4        | 1.5 trips         | 0.5 trips | Closed by emergency       | 1 trip                                                          | Converted to open area    | 1 trip                                |        | 18,000      |
| 2012              | FW22 and EA | 34    | 4        | 1 trip**          | 1 trip    | 0.5 trips                 | 1.5 trips                                                       | Closed (12/12/2012 by EA) | Closed by EA (trips converted to CA1) |        | 18,000      |
| 2013 <sup>1</sup> | FW24        | 33    | 2        | 118 trips***      | 182 trips | 116 trips                 | 210 trips                                                       | Closed                    | Closed                                |        | 13,000      |
| 2014 <sup>1</sup> | FW25        | 31    | 2        | Closed            | 197 trips | 116 trips                 | Closed                                                          | Closed                    | 313 trips****                         |        | 12,000      |
| 2015              | FW26        | 30.86 | 3 *****  | Closed            | Closed    | Closed                    | Merged into one Mid-Atlantic AA, but inshore part of ETA closed |                           |                                       |        | 17,000      |
| 2016              | FW27        | 34.55 | 3        | Closed            | Closed    | Closed ~                  | Merged into one MA AA, but inshore part of ETA closed           |                           |                                       |        | 17,000      |
| 2017              | FW28        | 30.41 | 4        | Closed            | 1         | 1                         | 1, plus another trip to ETA rotational area                     |                           |                                       |        | 18,000      |
| 2018              | FW29        | 24    | 6        | 1                 | Closed    | 2 NLS-W, 1 NLS-S          |                                                                 |                           | 2                                     |        | 18,000      |
| 2019              | FW30        | 24    | 7        | 1                 | Closed    | 3 in NLS-W                |                                                                 |                           | 3                                     |        | 18,000      |
| 2020              | FW32        | 24    | 5        | .5 FLEX           | 1         | .5 NLS-North, 1 NLS-South |                                                                 |                           | 2                                     |        | 18,000      |
| 2021              | FW33        | 24    | 4        | 856 GC trips, RSA | 1.5       | 1.5 NLS-South             |                                                                 |                           | 1                                     |        | 18,000      |
| 2022              | FW34        | 24    | 3        | GC Trips          | 2         | 1 NLS-South               |                                                                 |                           |                                       | Closed | 15,000      |
| 2023              | FW36        | 24    | 2        | Closed            | 2         | Closed                    |                                                                 | Open Bottom, ETA closed   |                                       | Closed | 12,000      |
| 2024              | FW38        | 20    | 3        | GC Trips          | 2         | Closed                    |                                                                 | Open Bottom               |                                       | 1      | 12,000      |

<sup>1</sup> Access area trips were allocated to FT LA vessels using a lottery. Numbers shown are total trips allocated per area (not per vessel).

\* FW18 also allowed vessels to exchange 2006 CAII and NLS trips for ETA 2007 trips

\*\*1 trip after emergency action May 2012 (157 vessels get initial trip per FW22 and 156 get CA1 trip converted from initial DMV trip)

\*\*\* FW25 then allows unused trips to be carried over to future year

\*\*\*\* Vessels given choice of Delmarva trip or 5 DAS

\*\*\*\*\* Vessels were not allocated trips in access areas, instead a poundage was allocated with a possession limit

~ NLS-N open to LAGC only

+ Information in this table prior to FY 2008 and before the implementation of limited access program in scallop fishery is available in FW30 or preceding scallop frameworks.